Project Name: Smart Inventory Management System

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Problem (The "Why")

- **Current Issue:** Inventory stock is currently difficult to monitor accurately because stock changes may not be recorded immediately or may rely on manual counting.
- **Impact:**
    - Stock information can become inaccurate.
    - Items may run out without being noticed.
    - Manual stock checking takes time.
    - Historical consumption is difficult to track.
    - Over-ordering or under-ordering may occur due to inaccurate stock information.
- **Our Approach:** We will develop an inventory management system that uses weight sensors to automatically detect changes in inventory quantity. Each inventory item will have a predefined average unit weight. When items are added or removed, the system detects the change in total weight and estimates the current quantity. The data will then be sent to the backend and displayed through an inventory dashboard.
- **Key Value:** The system reduces reliance on manual stock counting and provides near real-time inventory information. It also allows users to monitor consumption trends and identify when inventory needs to be replenished.

Core Features

- **Feature 1:** Automatic Weight-Based Stock Tracking, monitor inventory weight using a load cell and calculate the estimated quantity based on the configured unit weight
- **Feature 2:** Inventory Dashboard, display current inventory levels, item status, and recent stock changes.
- **Feature 3:** Inventory Usage History & Analytics, record stock changes and show historical consumption so users can identify frequently used items and estimate future inventory needs.
- **Feature 4:** Low Stock Alert, automatically notify the responsible user when inventory falls below its predefined minimum threshold.

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Clean Architecture
- **Design Patterns Used:**
  - Dependency Injection
  - Repository Pattern
  - Service Layer
- **Code Quality Goal:** We commit to maintaining a green Jenkins pipeline, writing unit tests to be tracked via Cobertura, and keeping Qodana static analysis warnings below 10 (Target: 0).

Identified Challenges & Risks

- **Risk 1:** Different items may have slightly different weights, and the measured weight can fluctuate due to sensor inaccuracies.
- **Risk 2:** If an individual item is very light, the weight difference may be too small for accurate detection.
- **Risk 3:** Synchronization between hardware and backend, network interruptions or sensor failures could cause the backend to receive outdated inventory information.